

Cover story **MEDICINE**

stacked with everything you need to make your body new again. A decade ago, not many knew how to harness their almost magical capability to heal the body. Now researchers are unlocking the mysteries these nano-sized cells store and are ushering in a revolution in the treatment of a range of debilitating diseases.

In Delhi, at the All India Institute of Medical Sciences (AIIMS), a steady stream of patients comes in for treatment of dilated cardiomyopathy, a condition in which the heart becomes weakened and enlarged. It cannot pump blood efficiently and one of its symptoms is a rapid pulse—twice the normal. Till now, short of having a heart transplant, it was among the most difficult cardiac disorders to treat. But for such patients there is new hope with AIIMS setting up a cardiovascular stem cell group that has begun, probably for the first time in the world, clinical trials to see whether the stem cells extracted from the bone marrow of patients, known as adult stem cells, could improve the heart's performance. The results are encouraging but Balram Bhargava, professor of cardiology and a co-principal investigator, says: "It's not magic yet. Still, in a field that has little options for patients, this is a definite improvement."

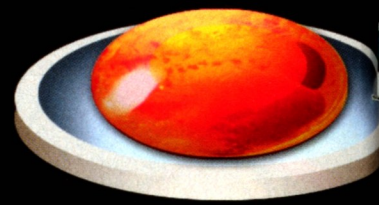
AIIMS is also conducting a major multi-centre trial to look at the role of stem cells in repairing tissue damaged during acute heart attacks. In many such cases, even bypass surgery doesn't seem to improve the functioning of the heart. Among the earliest beneficiaries of the wonders of such stem cell treatment is S. Kabilan, a former chief secretary of Assam. Four years ago he suffered a heart attack and underwent bypass surgery at AIIMS. When he heard about the new stem cell treatment which could regenerate damaged tissue, he readily consented to having it done. Since then the functioning of his heart has been regularly monitored and the ejection fraction—the amount of blood pumped out of a ventricle with each heartbeat—has shown considerable improvement. Says Kabilan: "I

THE BIOLOGICAL REPAIR SYSTEM

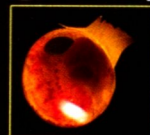
Though only a decade old, stem cell therapy holds the promise of radical and cost-effective treatment for medical disorders

① What are stem cells?

Stem cells are master cells that can divide and multiply rapidly into cells that give rise to the brain, heart, spine, the limbs, muscles, skin and everything else that constitutes the human body.

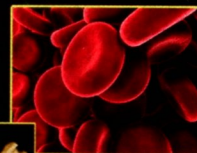


② Where they come from?



Embryonic Stem Cells

From embryos created during in-vitro fertilisation. These are the most potent and are used to produce many stem lines.



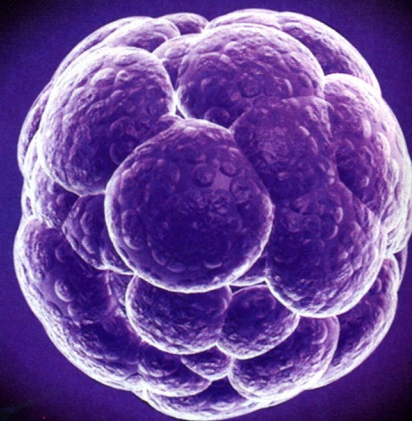
Umbilical Cord Cells

Taken from the umbilical cord at the time of delivery, they contain stem cells that can turn into bone, heart muscle and brain tissue.



Adult Stem Cells

Taken from the marrow of bones, they can be used to produce cells or tissue for a specific lineage such as cornea or heart muscle.



③ How is treatment done?

Stem cells are extracted in small quantities and injected into the body part or organ that is damaged or suffering from disorders.

④ What can it cure?

From diabetes, cardiovascular diseases, neurological disorders, burns and wounds, osteoarthritis, ocular and liver disorders, stem cells are being used for therapy.

Cardiovascular Diseases

Graphics by PRASHANT CHAUHAN